High-Performance $\star_{\mathbf{M}}$ t-SVD for Big Data Compression

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ILAS 2026 — May 19, 2026

Tensor Decomposition Landscape

Method	Optimality	HPC Ready
Matrix SVD	Exact (Eckart–Young)	✓
CP	None	✓
Tucker	$\sqrt{d} \times$ optimal	✓
Tensor Train	$\sqrt{d} \times$ optimal	✓
$\star_{\mathbf{M}}$ t-SVD	**Exact (Eckart–Young)**	✗

$\star_{\mathbf{M}}$ t-SVD: **exact optimality** — the only decomposition with no HPC implementation.

The Software Gap for $\star_M$ t-SVD

Existing

MATLAB & pure Python

- ✗ Single-threaded
- ✗ Limited parallelism
- ✗ GB–TB data infeasible
- ✗ Small experiments only

`pystarm`

C++ with Intel MKL + Python

- ✓ Shared-memory parallel
- ✓ Utilizes all available cores
- ✓ GB–TB datasets
- ✓ Python interface

1000× faster than currently available solutions on a **≈3 GB** Computational Fluid Dynamics (CFD) tensor

What We Tested On

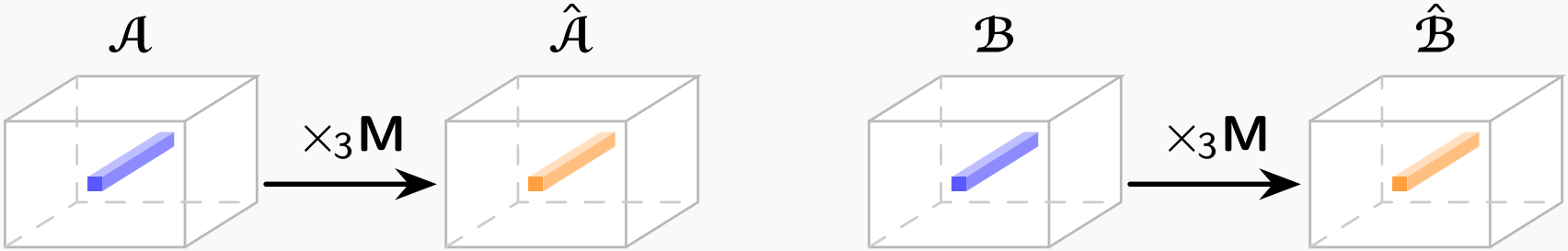
Dataset	Domain	Shape
ncep-air-6	Atmospheric reanalysis ($\approx 19.5\,\text{GB}$)	$73 \times 144 \times 17 \times 4 \times 365 \times 10$
cfd	Fluid dynamics ($\approx 3\,\text{GB}$)	$253 \times 253 \times 253 \times 4 \times 6$
xray	X-ray diffuse scattering ($\approx 366\,\text{MB}$)	$300 \times 400 \times 400$

System: ALCF Aurora node

- Dual-socket Intel Xeon Sapphire Rapids 9470C
- 52 cores per socket (104 cores total)
- Thread counts: 1–64

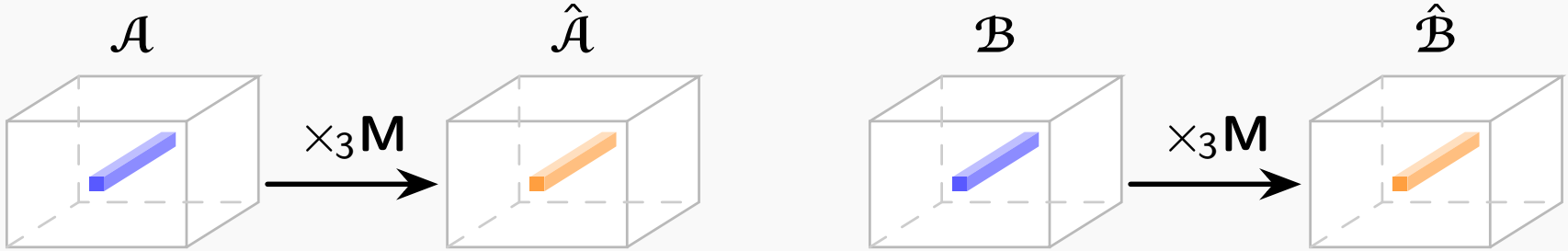
$\mathcal{A} \star_{\mathbf{M}} \mathcal{B}$: The $\star_{\mathbf{M}}$ Framework (Kilmer et al.2021)

Transform with invertible matrices

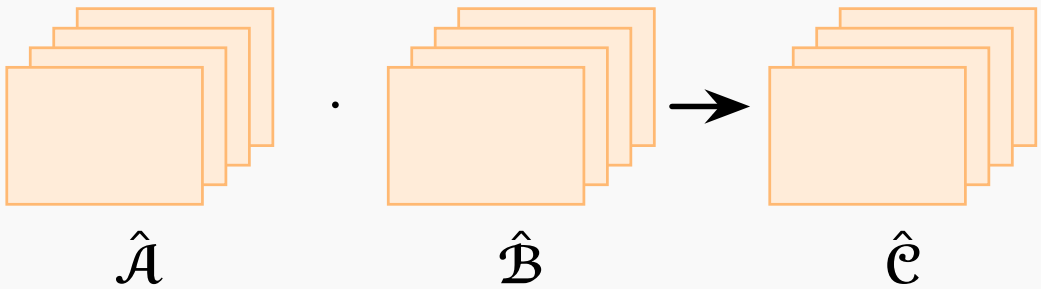


$\mathcal{A} \star_{\mathbf{M}} \mathcal{B}$: The $\star_{\mathbf{M}}$ Framework (Kilmer et al.2021)

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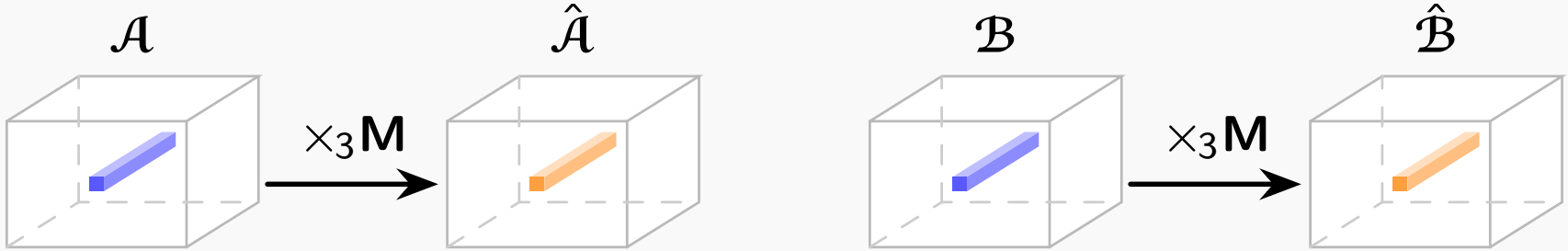


Slicewise-op

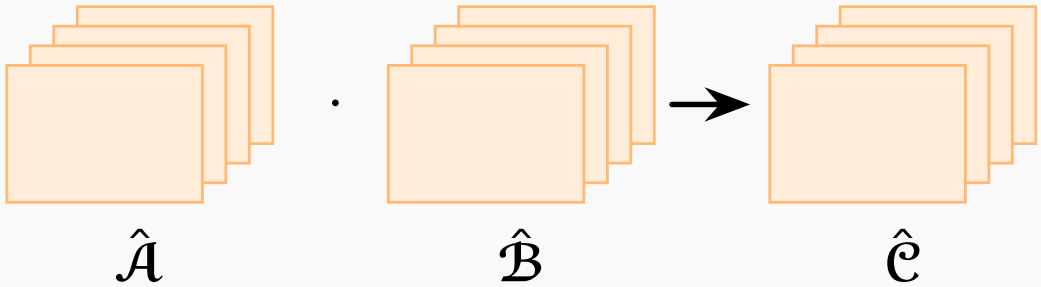


$\mathcal{A} \star_{\mathbf{M}} \mathcal{B}$: The $\star_{\mathbf{M}}$ Framework (Kilmer et al.2021)

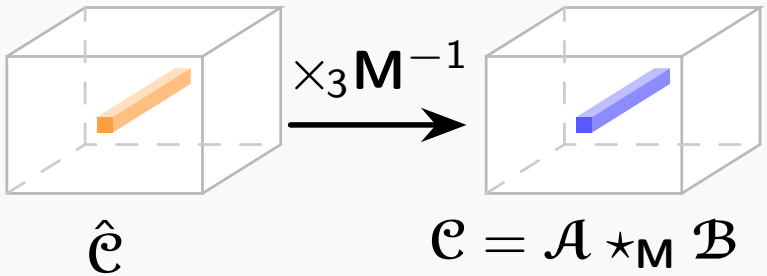
Transform with invertible matrices



Slicewise-op

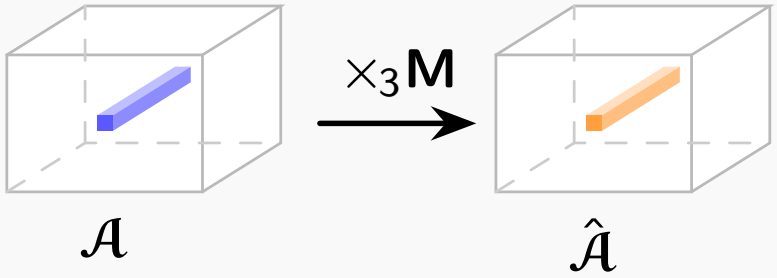


Back-transform

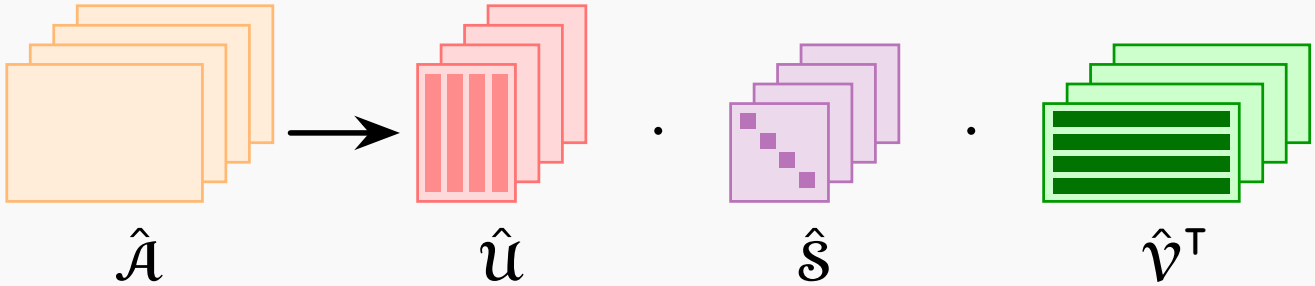


The $\star_{\mathbf{M}}$ t-SVD Algorithm

Transform



Slice-wise SVD

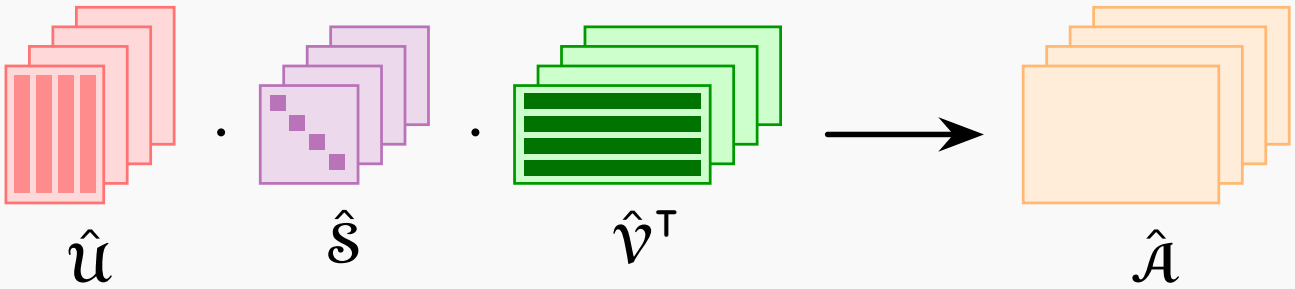


$$\star_{\mathbf{M}}\text{ t-SVD} = \text{TTM} + n\ \textbf{independent}\ \text{matrix SVDs}$$

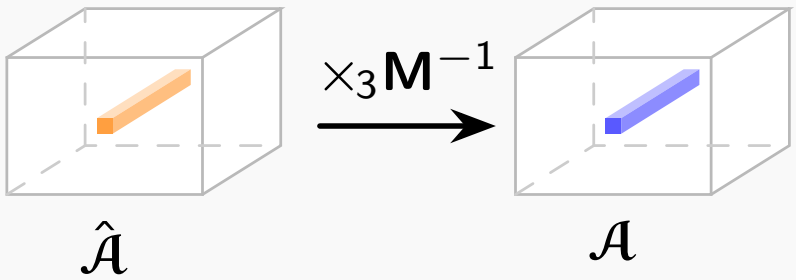
$$\hat{A}^{(i)} = \hat{U}^{(i)}\,\hat{S}^{(i)}\,\hat{V}^{(i)\mathsf{T}}, \quad i = 1, \ldots, n$$

$\star_{\mathbf{M}}$ t-SVD Reconstruction

Slice-wise multiply

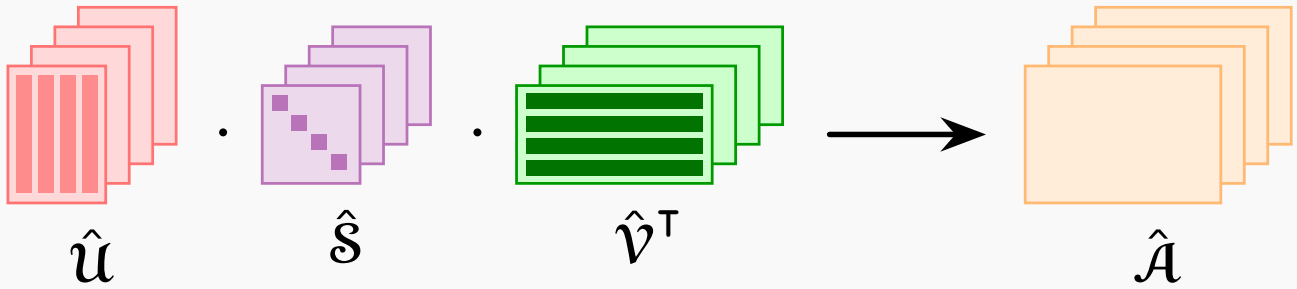


Back-transform

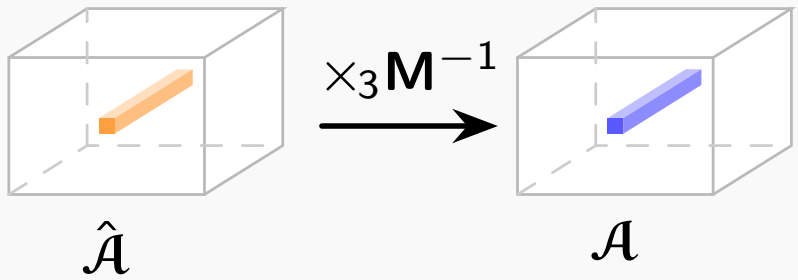


$\star_{\mathbf{M}}$ t-SVD Reconstruction

Slice-wise multiply



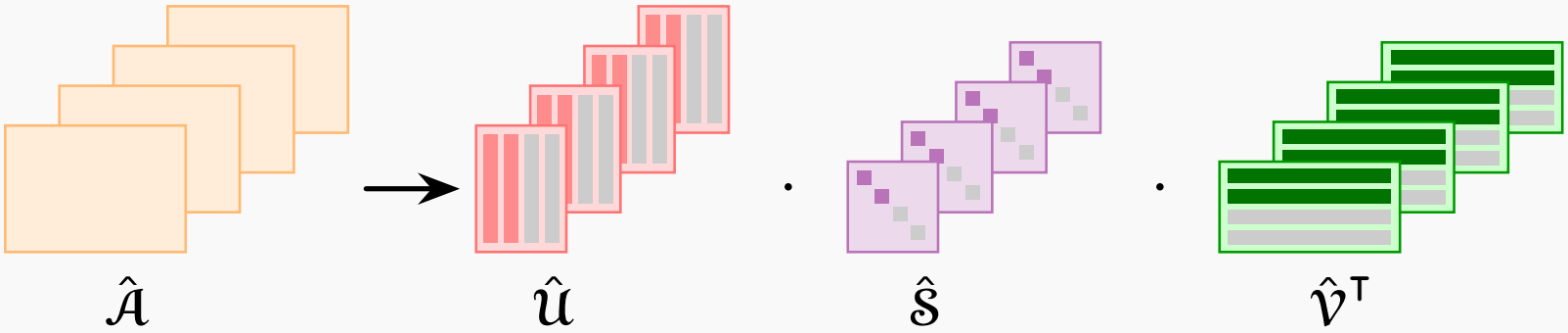
Back-transform



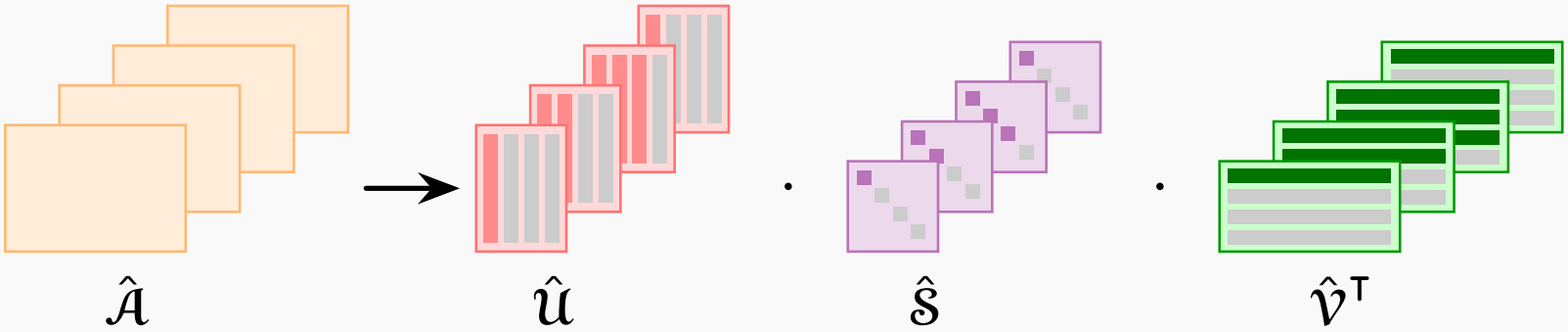
For d-way tensors: all modes beyond the 2nd are transformed — either mode by mode via individual TTMs, or equivalently via a single TTM with a Kronecker product transform. Slices are then treated identically.

Two Truncation Strategies

t-SVDM-I: truncate each slice to the same rank k (rank-specified)



t-SVDM-II: truncation at variable rank per slice (error-specified)



Variable rank truncation opens up scope for **better compression**

Compression Comparison: Climate Reanalysis Data

Tensor vs. Matrix

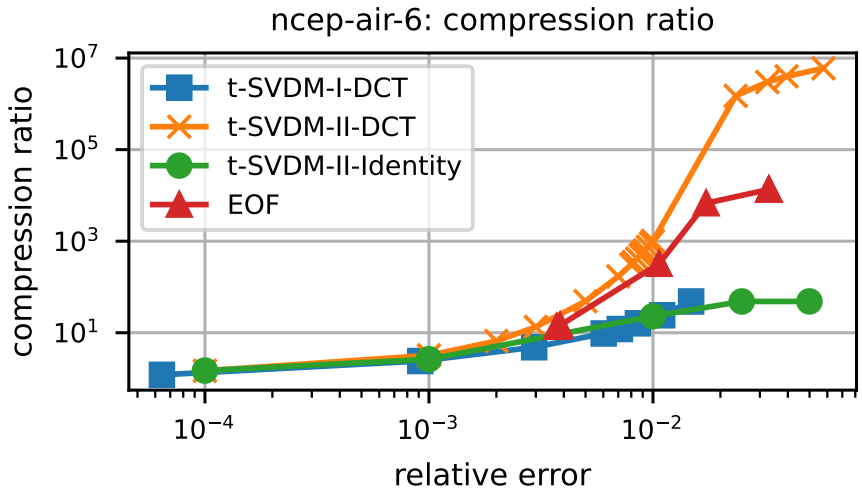
Tensor methods outperform EOF (Empirical Orthogonal Functions) compression used in climate science

Data

10 years of NCEP air temperature — 19.46 GB

$73 \times 144 \times 17 \times 4 \times 365 \times 10$

M: Discrete Cosine Transform (DCT)



Compression Comparison: Climate Reanalysis Data

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t-SVDM-II vs. t-SVDM-I

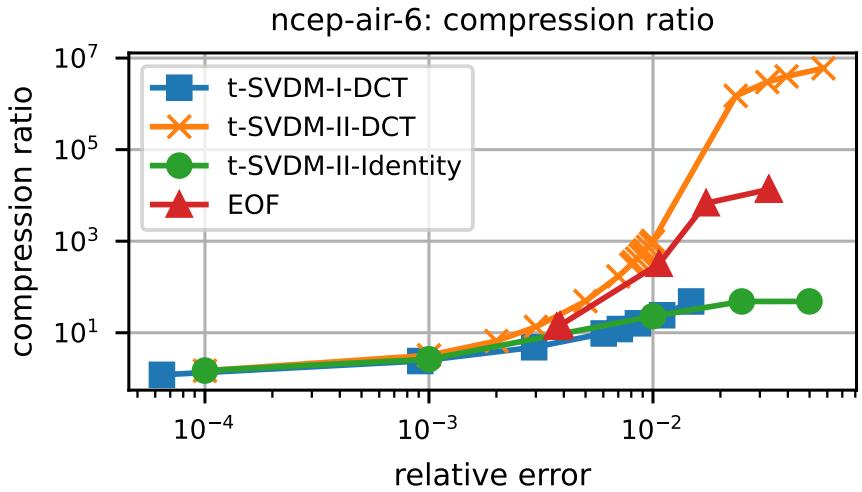
t-SVDM-II achieves higher compression at every error level

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Compression Comparison: Climate Reanalysis Data

Tensor vs. Matrix

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t-SVDM-II vs. t-SVDM-I

t-SVDM-II achieves higher compression at every error level

DCT vs. Identity

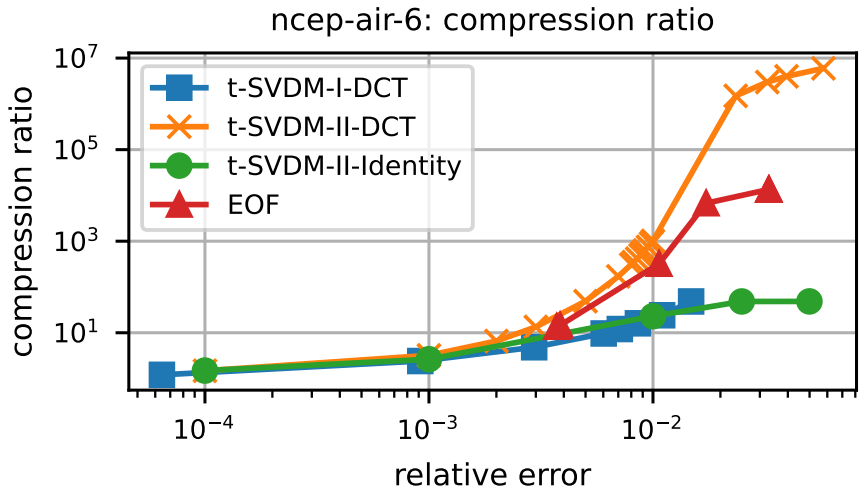
DCT concentrates energy into fewer singular values — better compression

Data

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M: Discrete Cosine Transform (DCT)

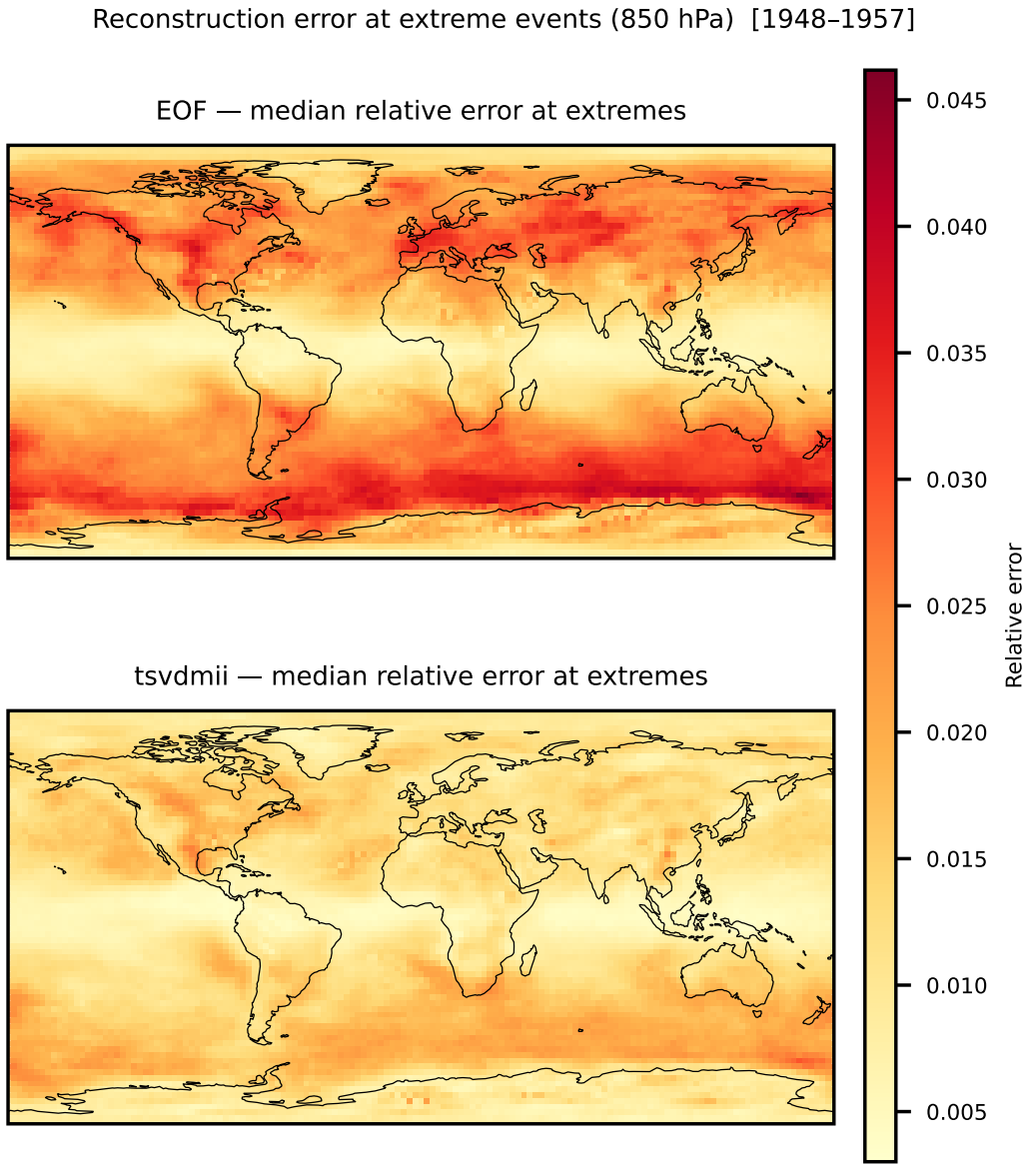


t-SVDM vs EOF: Extreme Event Preservation

Extremes
Top/bottom 0.5% at 850 hPa (surface air temperature)

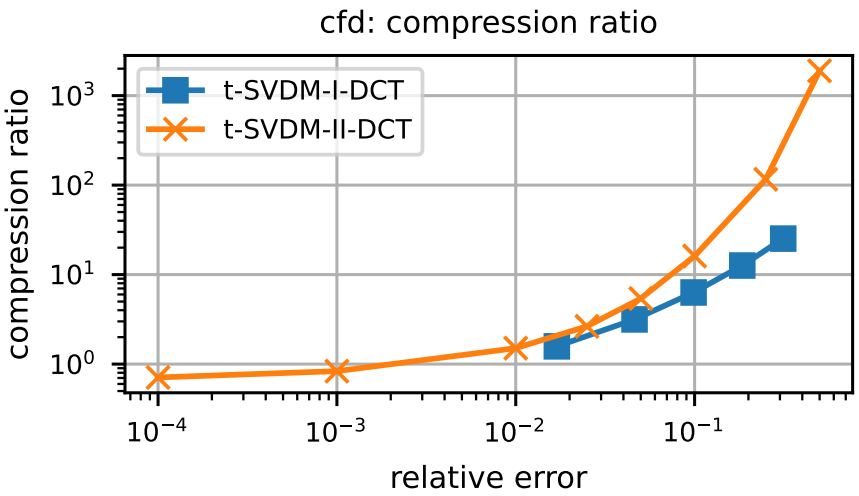
EOF smooths extremes
Tensor compression preserves them

Enabled by our work
19.46 GB — infeasible with existing implementations

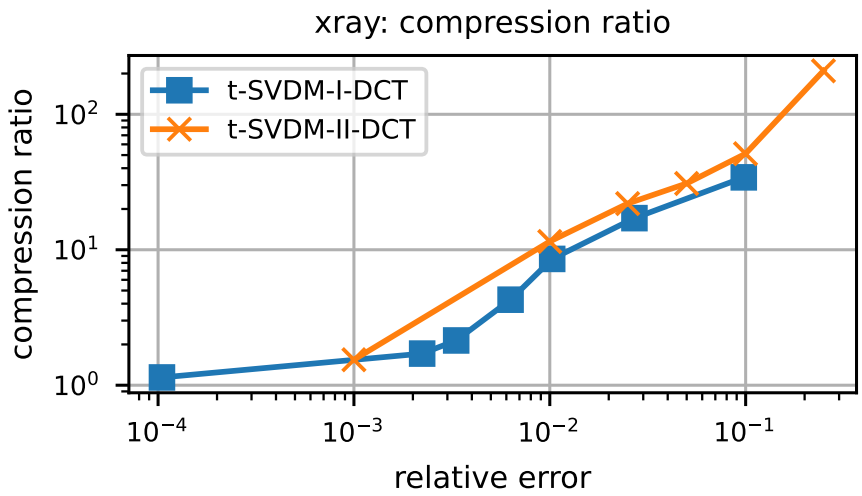


Compression Quality Across Scientific Domains

`cfd`: $253 \times 253 \times 253 \times 4 \times 6$ (≈ 3 GB)



`xray`: $300 \times 400 \times 400$ (≈ 366 MB)



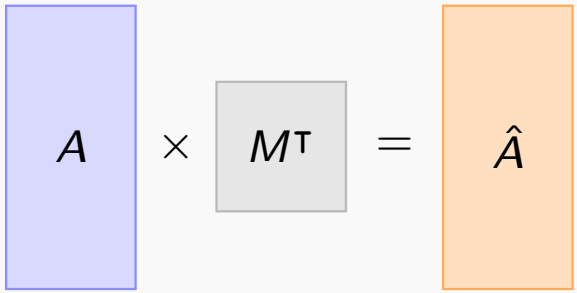
t-SVDM-II consistently outperforms t-SVDM-I across all datasets — the pattern holds broadly across scientific domains

t-SVDM Workhorse-1: TTM

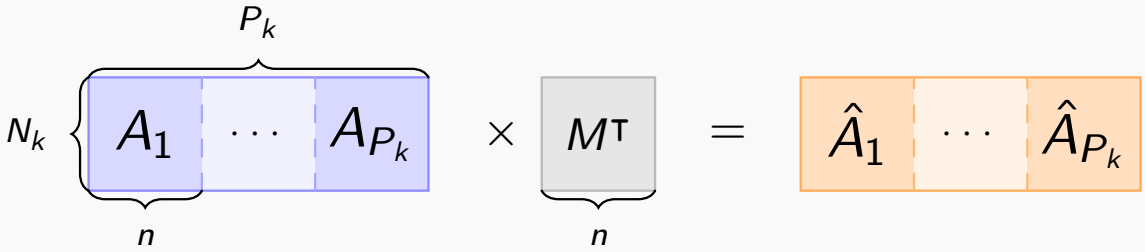
First mode: 1 GEMM



Last mode: 1 GEMM



Middle mode: strided GEMMs



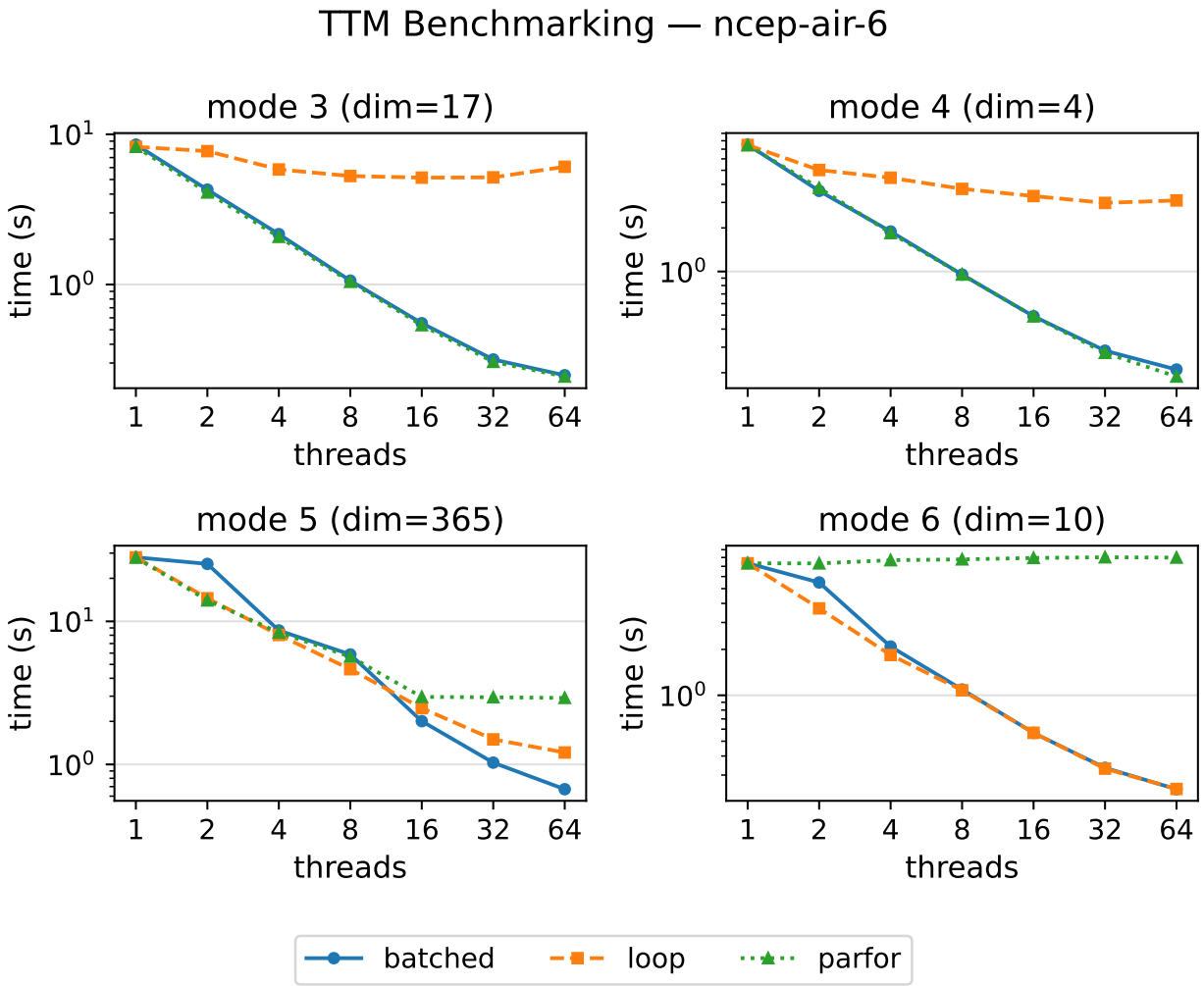
n: dimension of mode k; N_k: product of dims before mode k; P_k: product of dims after mode k

Possible designs

- *loop* — loop over multithreaded GEMM
- *parfor* — parallel-for over sequential GEMM
- *batched* — MKL batched strided GEMM

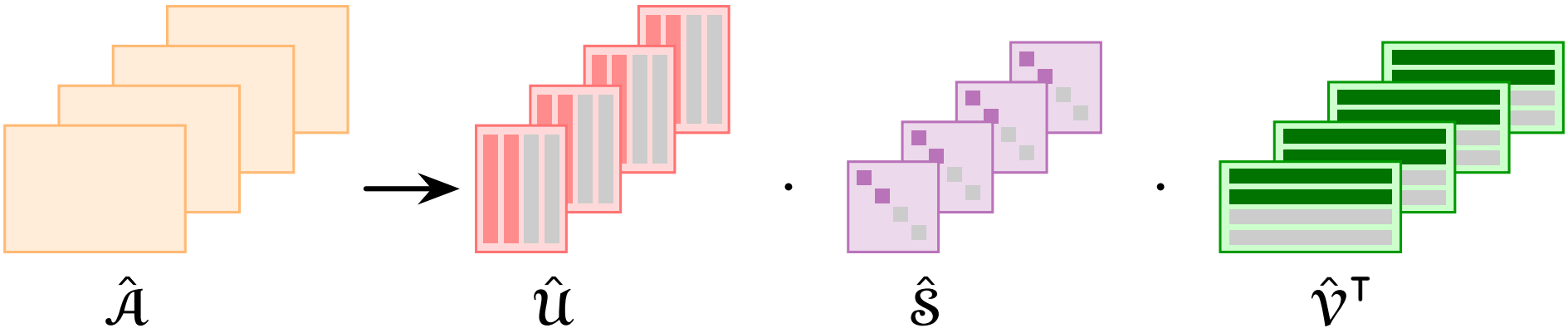
TTM Benchmark Results

`ncep-air-6`: $73 \times 144 \times 17 \times 4 \times 365 \times 10$



Batched GEMM wins across all modes and thread counts.

t-SVDM Workhorse-2: Slice-wise SVD

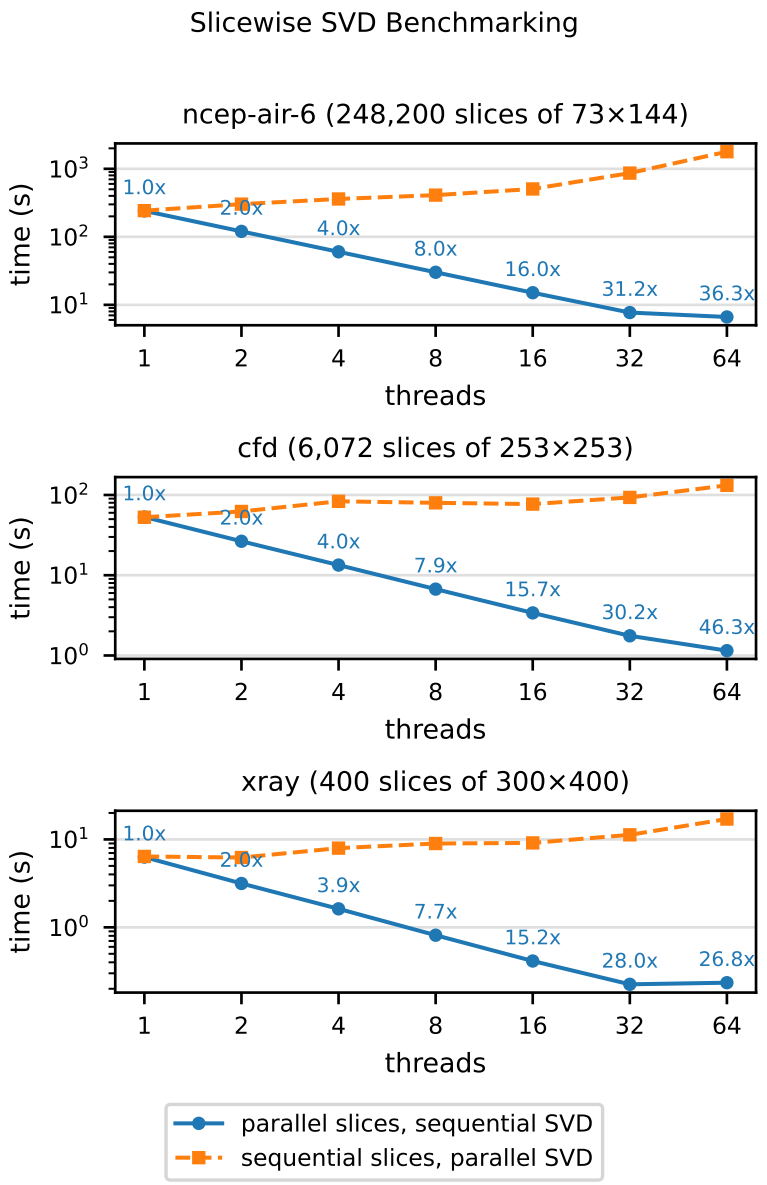


Possible designs

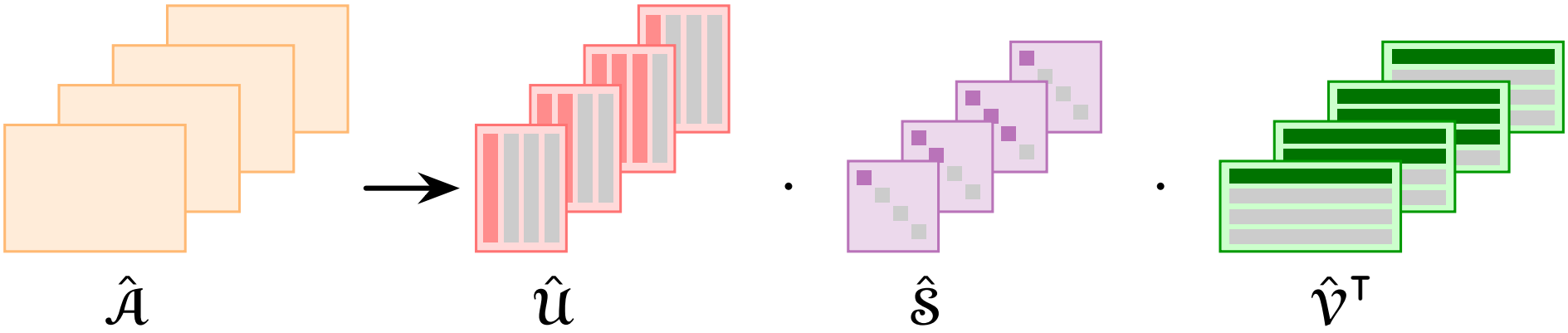
- **parallel slices, sequential SVD** — embarrassingly parallel, no dependencies between slices
- **sequential slices, parallel SVD**
- **a mix** — batched SVD, available in recent MKL but not as reliable

Slice-wise SVD Benchmark Results

- **parallel slices, sequential SVD** — near-linear scaling up to 64 threads
- **sequential slices, parallel SVD** — does not scale well, especially for small slices



t-SVDM-II Design Choices



one pass

single `dgesvd` call

truncate

- compute full SVD, drop below threshold

two pass: singular value computation + targeted vectors

compute-efficient

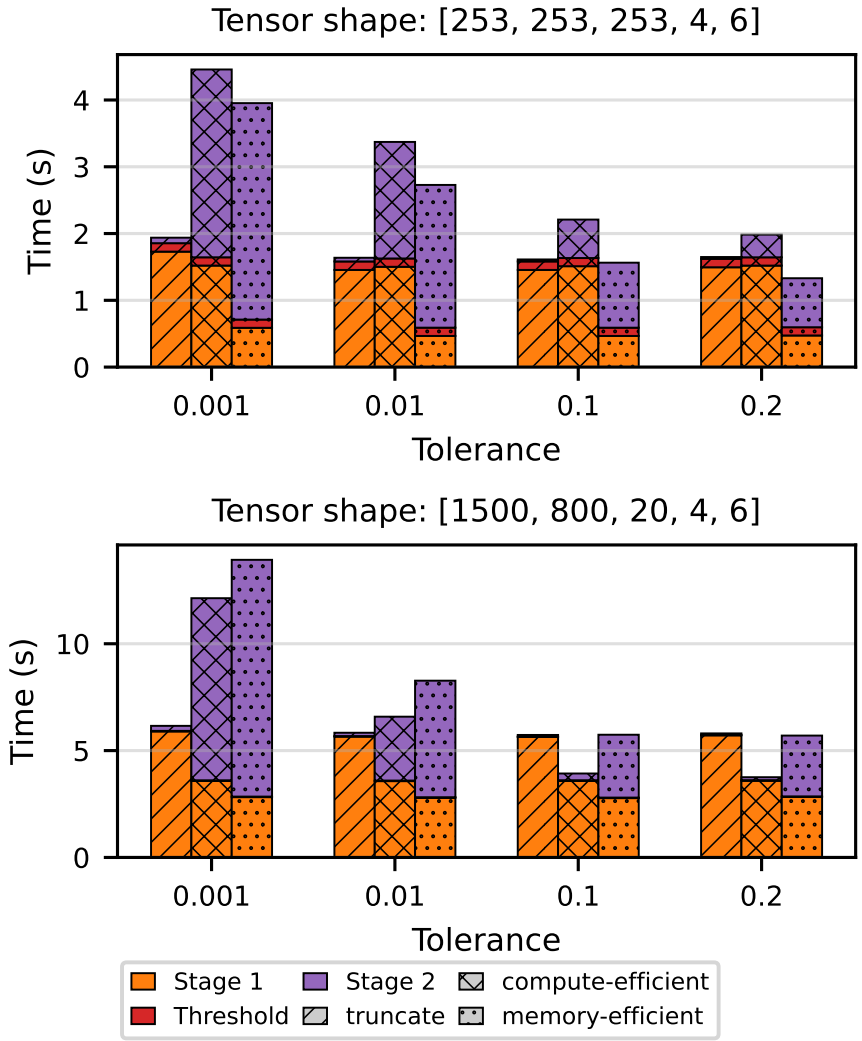
- caches QR and bidiagonal intermediates

memory-efficient

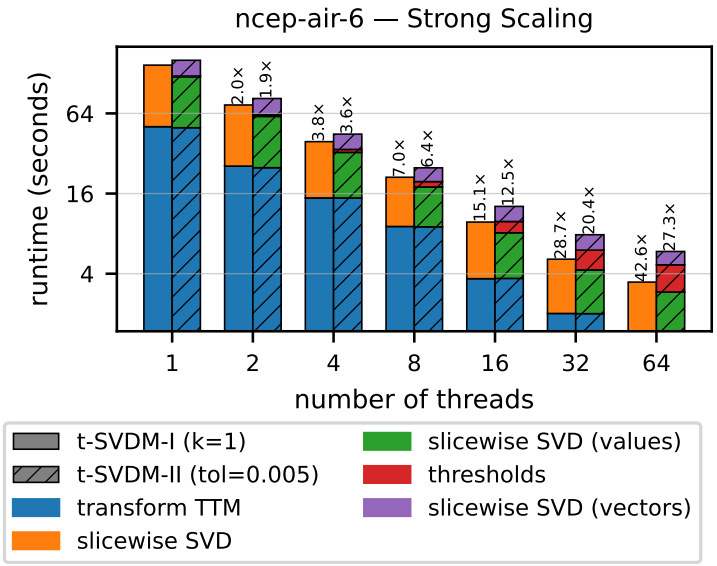
- no caching, least memory footprint

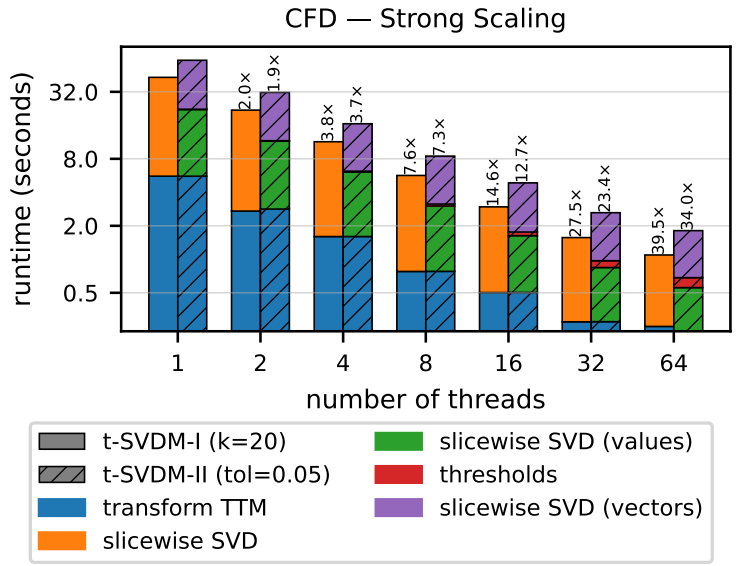
t-SVDM-II Benchmark Results

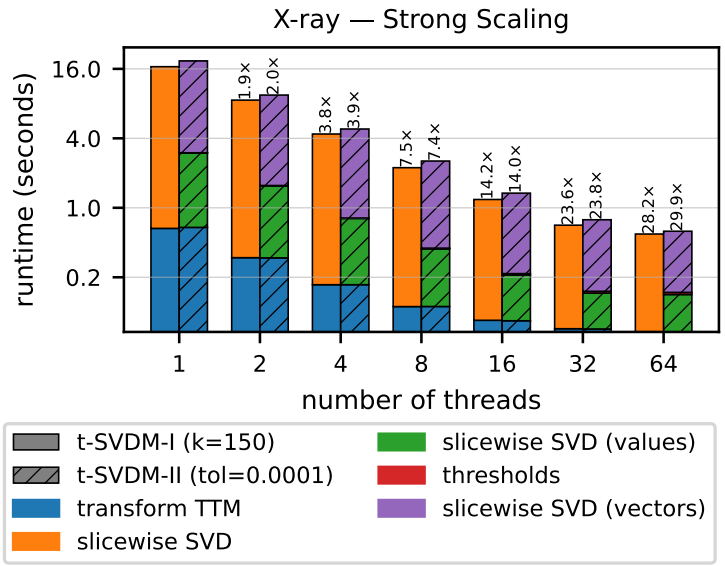
- **stage 1**: singular value computation, **stage 2**: singular vector computation
- no single winner — depends on tolerance and tensor shape
- truncate surprisingly competitive — `dgesvd` 8× faster than `dgesvdx`
- memory-efficient chosen for benchmarks



Strong Scaling Across All Datasets







Climate: 42× speedup

CFD: 39× speedup

X-ray: 30× speedup

Conclusions & Future Work

Conclusions

- ✓ First HPC implementation of $\star_{\mathbf{M}}$ t-SVD
- ✓ Good scaling to all available cores
- ✓ 1000× over currently available solutions
- ✓ Practical tool for large-scale compression

Future Directions

- GPU support
- Distributed-memory algorithms
- Operator transforms (FFT/DCT)

`pystarm` — coming soon.

http://arxiv.org/abs/2605.16058

References I

Misha E Kilmer, Lior Horesh, Haim Avron, and Elizabeth Newman. Tensor-tensor algebra for optimal representation and compression of multiway data. *Proceedings of the National Academy of Sciences*, 118 (28):e2015851118, 2021.